

HealthNet

HealthNet is a transformative initiative at the intersection of decentralized technologies and medical *open* innovation. By addressing systemic challenges in healthcare innovation and access, HealthNet pioneers a new model for creating, verifying, and disseminating *open source DIY* medical devices and software. Nondominium is an essential building block for HealthNet.



Young cystic fibrosis patient at Sainte Justine Hospital using the PEP Master device.

Vision and Purpose

HealthNet's vision is to establish trust and transparency in open source medical innovation while empowering global collaboration. By leveraging decentralized technologies, HealthNet creates an inclusive ecosystem where developers, researchers, medical professionals, and patients can work together to create high-quality, verifiable solutions that address critical healthcare needs. The platform generates incentives and motivations to sustain collaboration and ensures that open-source designs meet the highest standards of quality, safety, and provenance, fostering trust in these innovative solutions.

Origin and Context - PEP Master

HealthNet is developed within the framework of PEP Master, an open initiative focused on designing and disseminating affordable DIY medical devices for cystic fibrosis therapy. The platform builds on Sensorica's decade-long experience in *Open Value Networks* (OVNs), which promote collaborative and transparent innovation. Since its establishment in 2011, Sensorica has been a pioneer in *open source hardware* production, creating a framework for collaborative development and governance. Sensorica's infrastructure supports decentralized production networks by providing tools for transparent contribution tracking, accounting, and equitable distribution of benefits. This foundation enables HealthNet to operate as an ecosystem where stakeholders can contribute and benefit equitably from shared innovation. Sensorica's proven methodologies have facilitated partnerships with academic institutions, nonprofits, and industry leaders, demonstrating the effectiveness of open, peer-driven production systems.

The PEP Master is an *open venture* nurtured within Sensorica aiming to improve and disseminate an *open source DIY* therapy instrument for patients living with [cystic fibrosis](#). This instrument consists of a computer game and a hardware device (the game controller).

More precisely, this *open* venture is dedicated to help children living with cystic fibrosis, a rare genetic disorder that causes life-long chronic breathing problems. While pharmaceutical treatment alleviates symptoms and prevents potential complications, there is currently no cure for the disease. Additionally, the low numbers of cystic fibrosis cases worldwide (approx: 162,000)¹ acts as a counter-incentive for the private sector to fully invest. A radical new approach must be devised to provide dedicated, efficient and innovative measures.

The objective of our project is to provide an affordable mechanical technique used in managing the symptoms using positive expiratory pressure physiotherapy (PEP). We work in collaboration with Montreal Sainte Justine Hospital to test and validate this technology on patients, and potentially with the Western University to evaluate the technical aspects of the device against ISO standards. As this technology gets scientifically validated, we plan to challenge Health Canada to consider regulatory approval of this device and its associated therapeutic practice, considering the particularity of this solution, using the *commons-based peer production* model, i.e. being designed by a network of *open source software* and *hardware* developers, destined for distributed fabrication.

PEP Master was initiated by Breathing Games in collaboration with Sensorica, and was brought to the attention of medical professionals research labs. It received a 100,000 CAD grant in 2024 in collaboration with Dr. Sze Man (from Sainte Justine Hospital in Montreal) and Ludociels (non-profit organization). For more on Breathing Games please visit the [website](#).

The PEP Master Partnerships

PEP Master serves as HealthNet’s initial use case, illustrating the platform’s potential to deliver tangible healthcare solutions. The initiative highlights how decentralized technologies and collaborative networks can address pressing healthcare challenges. Key partnerships and their contributions include:

	Sensorica Open innovation and peer production network	Plays a central role in open source hardware development, the PEP Master electronic pressure sensor. Provides the organizational structure for open source hardware innovation
	Breathing Games Open innovation and peer production network	Plays a central role in software development, the PEP Master computer games. Provides the organizational structure for open source software innovation.

	Envienta Blockchain development	Web3 infrastructure development.
	Ludociels	Plays a central role in UX and UI design. Provides custodial services to Sensorica, and plays the role of open network-to-institution interface.
	Sainte Justine Hospital	Plays a central role in clinically validating the PEP Master device, ensuring its safety and efficacy for cystic fibrosis therapy. Develops usage protocols and monitors patient outcomes to refine device effectiveness.
	Western University	Provides technical validation of the device's design and compliance with ISO manufacturing standards. Ensures that the device meets engineering and regulatory requirements for global scalability.
	Axelys	Technology transfer intermediary, providing guarantees that the PEP Master device remains open source.
	Health Canada	Engages with the project to explore new regulatory pathways for certifying open-source medical devices developed outside traditional accredited supply chains.

By integrating these partnerships, PEP Master demonstrates how HealthNet bridges the gap between decentralized innovation and institutional trust, creating a replicable model for future initiatives in healthcare and beyond.

The HealthNet Platform

At the base layer, HealthNet is a P2P platform that facilitates the authentication of *open source hardware* designs and *open source software*. It will provide new tools for developers in the *open source* communities to catalog their creations and make them accessible to the public. In addition to authentication, HealthNet will offer tools and methods for version tracking, allowing *open source* developers to fork and remix existing creations (similar to Git) while adhering to strict protocols for maintaining quality, safety, provenance and authenticity. These features enable developers from various *open source* communities and networks to collaborate seamlessly across organizational and institutional boundaries, to innovate and improve designs, reducing the problem of unproductive proliferation (forking) and competitive dynamics.

Another requirement for HealthNet is to provide economic incentives and motivation for *open source hardware* and *software* developers, and to ensure the sustainability of the platform.

As already mentioned, HealthNet is developed in the context of [PEP Master](#), an *open* initiative to develop *open source and DIY* medical devices and associated software. We are planning to leverage the patient-doctor relationship and the requirement for high levels of trust in the medical sector to implement measures that ensure financial rewards for developers, using existing web3 primitives. This is part of what we call the [Trust Economy scheme](#).

Regulatory approval

HealthNet will be an ecosystem service in the medical device sector, where ensuring the quality¹, safety², provenance and authenticity of all available designs is of paramount importance. Regulatory approval for medical devices anchors trust in the proposed solutions in standardized methods and processes that are embodied by traditional medical device companies. *Open source DIY* medical devices are designed by *open source* communities and *open* networks, and are fabricated by local agents, which operate outside of the currently regulated supply chain. How can we rebuild trust in a distributed system of production and dissemination, as an alternative to the traditional accredited supply chain? P2P technologies provide new solutions to this trust issue, allowing for a wider intervention in the medical field.

In a few words, as new *open source* medical solutions emerge within *open source* communities, they undergo rigorous scientific evaluation in collaboration with academic institutions. Most academic institutions already engage with *open source hardware* and *software*. A feedback loop is established between the group of freelance designers and academic research labs to mature these solutions to their required level of safety and security. Tight collaboration between *open source* communities and traditional academic institutions has been practiced by [Sensorica](#) since 2011, which is incubating HealthNet.

The satisfactory design of an *open source* solution is published in peer review scientific journals, which will constitute the anchor of trust and pave the way to certified use. Patients, the primary users of this *open source* solution, can seek medical assistance for its use, while medical professionals, including doctors and healthcare workers, can have confidence in the quality and effectiveness of the solution, as attested by the scientific publications. Cryptographic methods can be designed to build this chain of trust.

The initiative collaborates with researchers from Montreal Sainte-Justine Hospital to clinically test and validate the device with patients, and intends to work with researchers at Western

¹ Quality is insured by certification or regulatory compliance of these creations.

² Safe means distributed (better than decentralized), secure (encrypted, authentication, temper-proof, end-to-end) and private (possibility to create a private network).

University to evaluate the device's technical compliance with ISO standards. As the technology nears maturity, PEP Master engages Health Canada to explore paths to regulatory approval, considering the unique aspects of how the technology is developed and disseminated.

HealthNet plays a key role in PEP Master's negotiations with Health Canada, by providing a new path to build trust in open source DIY solutions that are created and disseminated outside of the traditional accredited medical supply chain. These new channels of medical device production offer significant benefits in areas where the private sector has minimal or no incentives to invest, such as rare medical conditions.

In addition, HealthNet, like PEP Master, leverages cutting-edge collaborative tools and methodologies, such as decentralized project stewardship and decentralized finance (DeFi). Both projects are nested within the Sensorica OVN and benefit from its collaborative infrastructure.